Analog / Digital / Pins

digitalWrite - turns on a particular pin to HIGH or LOW

digitalWrite(8, HIGH)

pin value

HIGH = 5V → 1

LOW = 0V → 0

digitalRead - read the signal at a pin, and return HIGH or LOW

- if the signal is > 2.5V → HIGH

  ≤ 2.5V → LOW

(uses an analog-to-digital converter ADC)

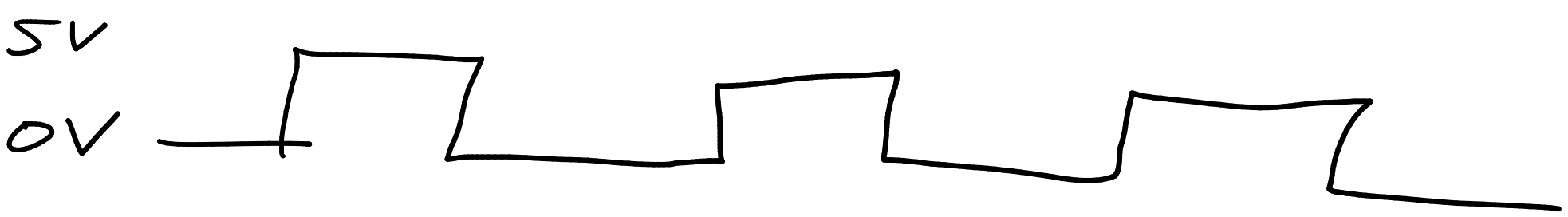
analogRead (on the analog pins only)

- reads a signal, and returns a value 0 ⟷ 1023, depending on voltage

0V ⟷ 0

5V ⟷ 1023

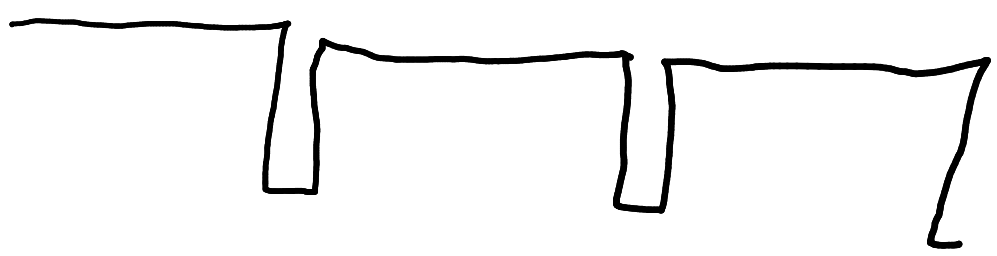
analogWrite - emits alternating 0V and 5V pulses



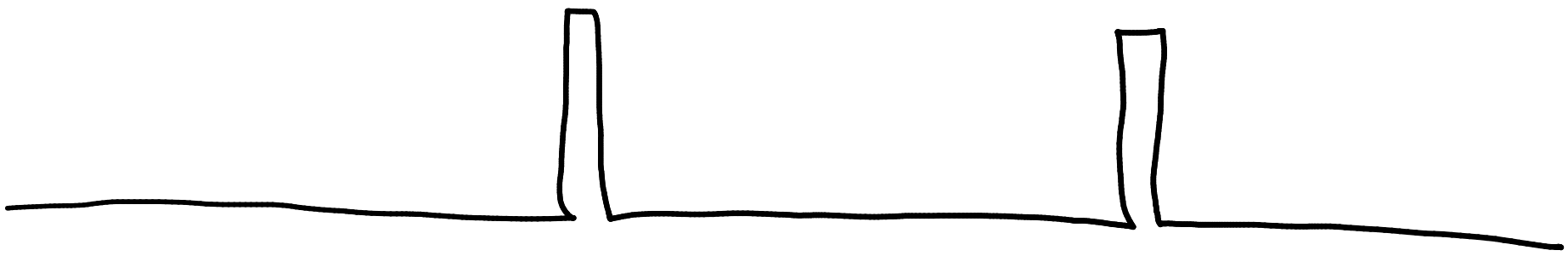
the number you supply is the fraction of the time the pulses are high

analogWrite(9, 240)

pin (close to 255



analogWrite(9, 20)



process: pulse width management (PWM)

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which pins do what?

Digital pins: (D)0 - (D)53
digitalWrite
digitalRead

Caveats:
pin 13 connected to built-in LED
(don't use 13 unless for that)

pins 0 and 1 overlap w/ laptop communication (so avoid)

Only some of the digital pins can do PWM (analogWrite)

pins D2 - D13, D44 - D46

(avoid D13 except for built-in LED)

Analog pins (A0 - 15)

- can read analog signals and give you a number
- analogRead works
- you can also use digitalRead/digitalWrite, but ... why?

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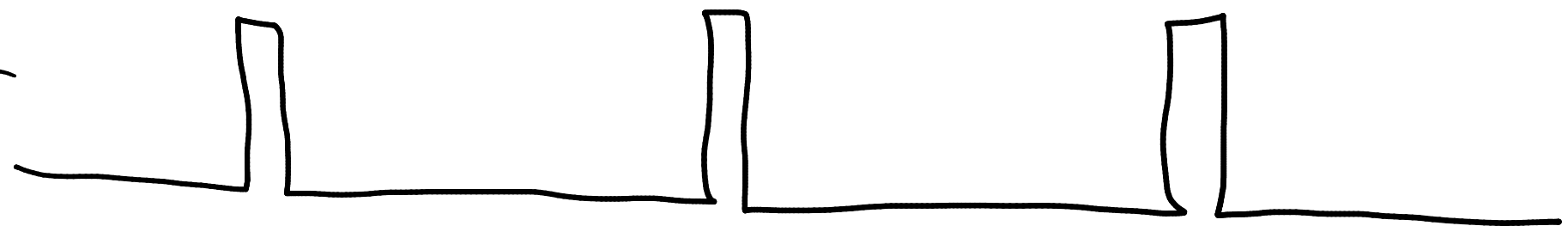
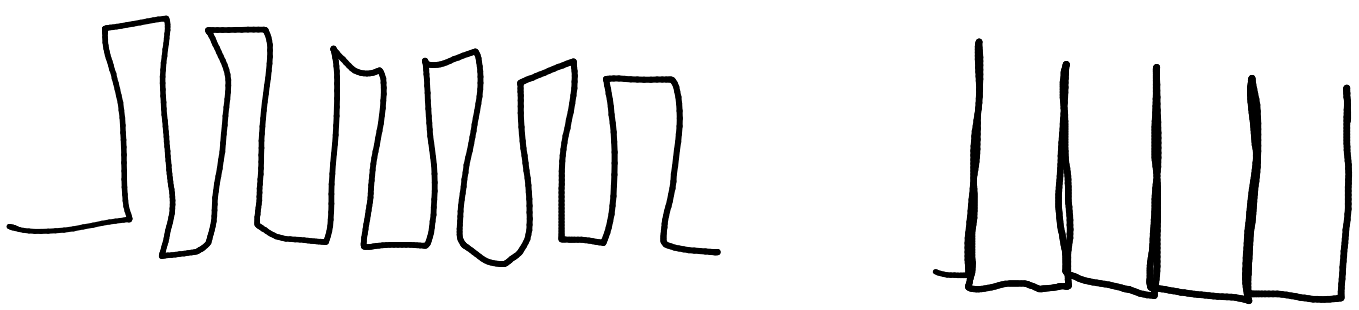
tone needs to output at a paticular frequency

Frequency - how often a pulse happens

A low freq sound



A high freq sound

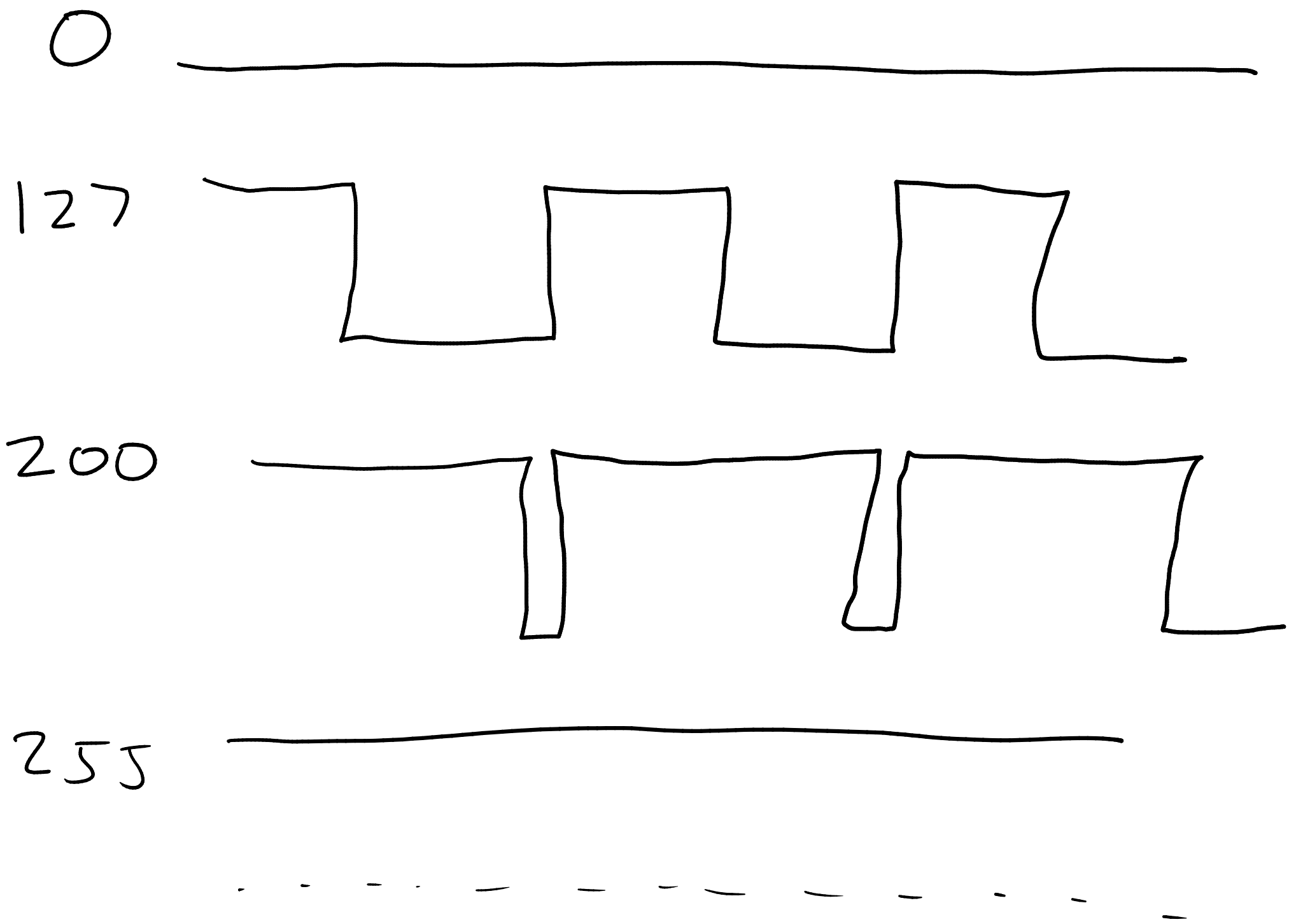


PWM controls how much of the time signal is high

- different from how often it cycles from low → high → low (tone does that)

<u>tone</u> does this using Arduino
timers
Go clocks in Arduino it can use
to manage the signal

PWM

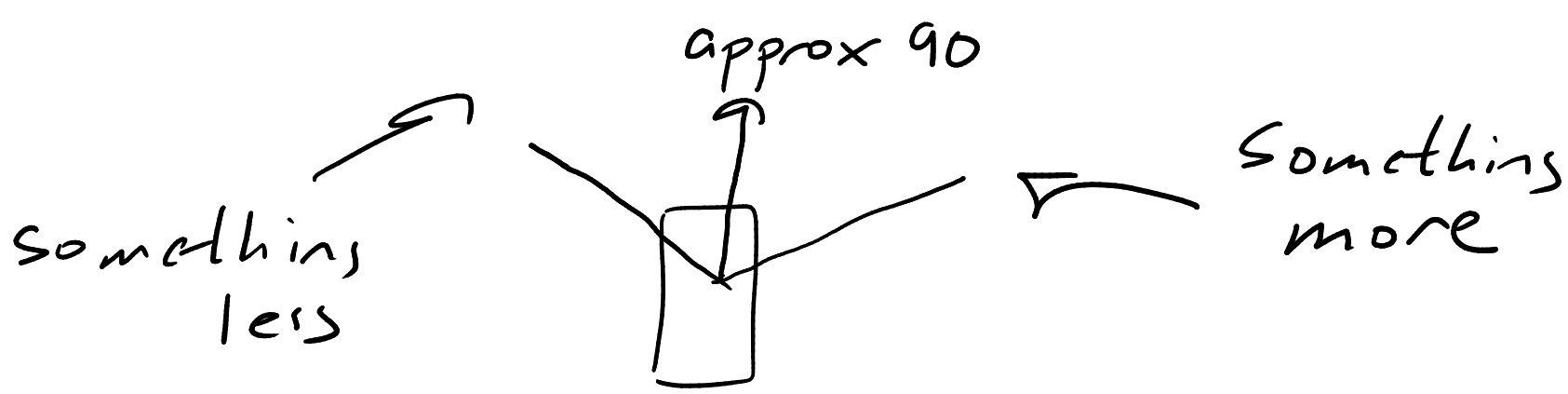


We've got servos!

(like a motor, but electronically controlled)

We have two types:

positional - value you give it determines position



More about pins.

It turns out that the servo library sends pulses to the servos to control them.

It is not the same kind as PWM.

Kind of like tone in that servo library sends its own kind of pulse signal.

Like tone library, it uses built-in timers to do this.

There are 6 built-in timers on the Arduino mega.

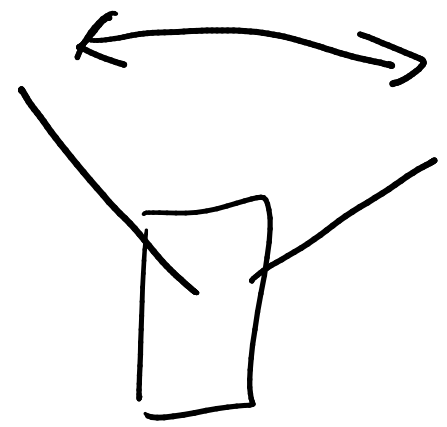
Our servo library

Servos can go on any pin, but the use timers (see charts in studio), which can collide w/ other things that use timers

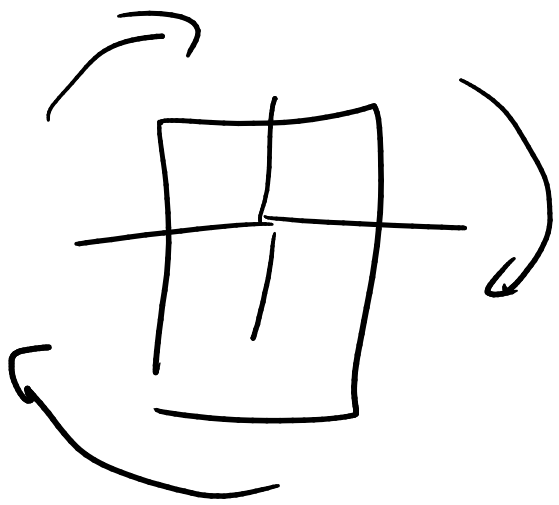
- PWM/analogWrite
- tone (timer 2)

Two kinds of servos:

- positional



- continuous rotation (CR)



spins

value you give it determines speed and direction

myServo.write(90) is approx not moving

ish

larger → spin clockwise
smaller → spin counter-clockwise

plugging servos:

3 wires - colors vary

- orange/red/-ish - voltage
- brown/black-ish - ground
- directly match 3-pins on our Arduino